

MAE 145: Planning and Estimation

Homework #3

Assigned Apr 16. Due on Apr 23

Notice:

Please upload 2 files for this homework. The first file should refer to the written part of the homework. The second file should consist of the program solving the last problem (see the template on canvas). Include in your written and electronic assignments the names of the other students you may have collaborated with to do the homework problem. The Collaboration Policy for this course is detailed in the syllabus.

Homework exercises:

1. Make use of a search algorithm to find the shortest path from node α to θ . The graph, with cost edges, is given in Figure 1. Write several snapshots of the result, as your queue changes under the algorithm.

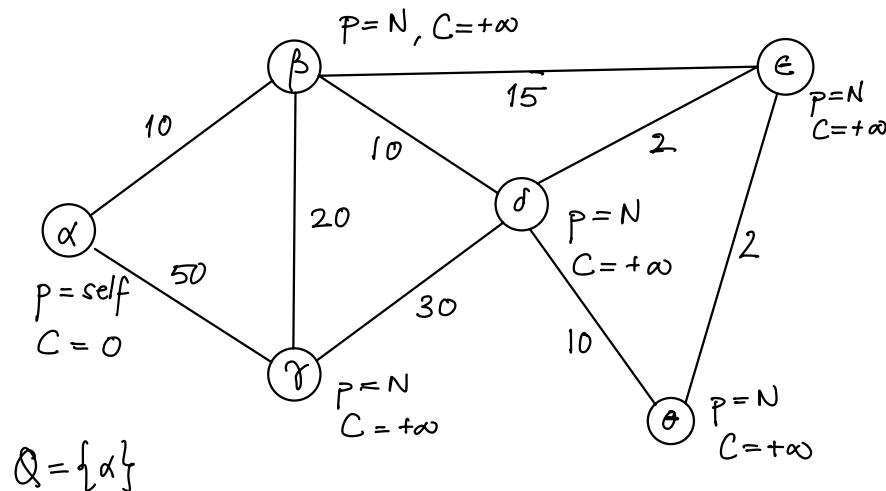


Figure 1: Input graph for problem 1. The value p represents the parent of a node, with $p = N$ none assigned.

2. Exercise E3.2, from the lecture notes by Bullo and Smith.
3. Exercise E3.5 from the lecture notes by Bullo and Smith.
4. Programming Exercise E2.8, from the lecture notes by Bullo and Smith. See the program template on canvas and test case (next page.)

Note that there is a test case that are part of the script (after the line `if __main__ == 'main':`). Feel free to test your code in as many graph examples as you like, as this will help you debug your algorithm. In addition:

- (a) The output of `computeBFStree(AdjTable,start)` should be a list (rather than a math vector). The entries of the list should be a sequence ordered by the time they were assigned a parent. For example, suppose we have nodes 'a', 'b', 'c', 'd', 'e', 'f' as part of our graph, with `start='a'`. Then suppose that the BFS assigns in order `parent(a) = a`, `parent(b) = a`, `parent(d) = a`, `parent(f) = d`, `parent(c) = d`, `parent(e) = f`. Then the BFS output of this result is ['a', 'b', 'd', 'f', 'c', 'e'].
- (b) The output of `computeBFSpth(AdjTable,start,goal)` should be the list consisting of the sequence of nodes from `start` to `goal` resulting from the BFS search. In the example above, if `goal = 'c'`, then the output path is ['a', 'd', 'c']. You can use the output of `computeBFStree(AdjTable,start)` in order to extract the path.